

26) (a) 1)

$$\forall x (P(x) \wedge Q(x))$$



$$\forall x P(x) \wedge \forall x Q(x)$$

let y be arbitrary

$$\text{Esso } P(y) \wedge Q(y) \quad \text{--- These can be treated as 2 separate pieces}$$

as $P(y)$ & $Q(y)$ was arbitrary $\rightarrow \forall x P(x)$
 similar for Q .

$$\text{As } P(y) \wedge Q(y) \text{ with arbitrary } y \rightarrow \forall x P(x) \wedge \forall y P(y)$$

$$\forall x P(x) \wedge \forall x Q(x) \longrightarrow \forall x (P(x) \wedge Q(x))$$

let y be arbitrary, then

$$P(y), Q(y) \text{ so } P(y) \wedge Q(y)$$

Since y was arbitrary, so

$$\forall x (P(x) \wedge Q(x))$$

$$2) A \subseteq B \wedge A \subseteq C$$

$$A \subseteq B$$

$$A \subseteq C$$

$$\forall x [x \in A \rightarrow x \in B] \quad \text{--- ①}$$

$$\forall x [x \in A \rightarrow x \in C] \quad \text{--- ②}$$

$$x \in A \quad \text{③}$$

$$x \in B$$

$$x \in C$$

$$\text{so } x \in B \wedge x \in C$$

$$A \subseteq B \cap C$$

$$\forall x [x \in A \rightarrow x \in B \cap C]$$

let x be arbitrary

$$x \in B \cap C$$

$$x \in B \wedge x \in C$$

modus ponens
 ① ② ③

$$3) A \subseteq B$$

$$\forall x [x \in A \rightarrow x \in B] \text{--- (1)}$$

$$\forall C [C \cap B \subseteq C \cap A]$$

let C be arbitrary set

$$C \cap B \subseteq C \cap A$$

$$\forall x [x \in C \cap B \rightarrow x \in C \cap A]$$

let x be arbitrary

$$x \in C \cap B$$

$$x \in C \cap A$$

$$x \in C \wedge x \notin B \text{--- (2)}$$

$$x \in C \wedge x \notin A$$

$$x \notin A \therefore \text{modus tollens (1) \& (2)}$$

$$\text{so } x \in A \wedge x \notin C$$

$$4) A \subseteq B$$

$$A \not\subseteq C$$

$$B \not\subseteq C$$

$$\forall x [x \in A \rightarrow x \in B] \text{--- (1)}$$

$$\exists x (x \in A \wedge x \notin C) \text{--- (2)}$$

$$\exists x (x \in B \wedge x \notin C)$$

$$\downarrow$$

$$x_0 \in A \wedge x_0 \notin C \therefore \text{existential inst (2)}$$

$$x_0 \in B \therefore \text{modus ponens (1)}$$

$$\text{so let } x = x_0 \text{ then } x_0 \in B \wedge x_0 \notin C$$

$$B \not\subseteq C$$

$$5) A \subseteq B \cap C$$

$$A \neq \emptyset$$

$$B \not\subseteq C$$

$$\exists x (x \in B \wedge x \notin C)$$

$$\forall x (x \in A \rightarrow x \in B \cap C) \text{--- (1)}$$

$$A \neq \emptyset \rightarrow \exists a \in A \text{--- (2)}$$

$a \in B \cap C$: Modus Ponens D20

$a \in B \wedge a \notin C$

So let $x = a$ (this type of element)

$$b) A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$$

let $x \in A \setminus (B \cap C)$ & x be arbitrary

$x \in A \setminus (B \cap C)$ iff $x \in A \wedge x \notin B \cap C$: def

iff $x \in A \wedge (x \notin B \vee x \notin C)$: def

iff $x \in A \wedge (x \notin B \vee x \notin C)$: de Morgan

iff $(x \in A \wedge x \notin B) \vee (x \in A \wedge x \notin C)$: negation

iff $(x \in A \setminus B) \vee (x \in A \setminus C)$: Association

iff $x \in (A \setminus B) \cup (A \setminus C)$: def

iff $x \in (A \setminus B) \cup (A \setminus C)$: def

Since x was arbitrary $\rightarrow$ Hence proved

$\rightarrow P(A \cap B) = P(A) \cap P(B)$ let A & B be arbitrary sets

$P(A \cap B) \rightarrow$

$P(A) \cap P(B)$

let x be arbitrary

$x \in P(A \cap B)$

$x \subseteq A \cap B$

~~let~~ $\forall y [y \in x \rightarrow y \in A \cap B]$

let y be arbitrary

$y \in x$

$y \in A \wedge y \in B$

$\rightarrow$ since y was arbitrary

so $x \subseteq A$

or $x \subseteq P(A)$

($\therefore$ similar for B)

Since x was arbitrary so $P(A) \cap P(B)$

Note that $\forall y [y \in x \rightarrow (y \in A \wedge y \in B)]$
 simplifies to $\forall y [y \in x \vee (y \in A \wedge y \in B)]$
 $\forall y [(y \in x \vee y \in A) \wedge (y \in x \vee y \in B)]$
 $\forall y [(y \in x \rightarrow y \in A) \wedge (y \in x \rightarrow y \in B)]$

These can be treated as 2 givens which implies
 that $x \subseteq A \wedge x \subseteq B$.

$x \in P(A) \wedge x \in P(B) \rightarrow P(A \cap B) \rightarrow$
 As x was arbitrary $\Rightarrow P(A) \cap P(B)$

~~$A \subseteq B \text{ iff } P(A) \subseteq P(B)$~~

Now $P(A) \cap P(B) \rightarrow P(A \cap B)$

Let A, B be arbitrary sets
 Let $x \in P(A) \cap P(B)$ x is arbitrary
 $x \in P(A) \wedge x \in P(B)$

$x \subseteq A \wedge x \subseteq B$

Let y be arbitrary

$\forall y [y \in x \rightarrow y \in A]$

$\forall y [y \in x \rightarrow y \in B]$

Suppose $y \in x$ then $y \in A$

$\& y \in B \rightarrow y \in A \cap B$

Since y was arbitrary

so $x \subseteq A \cap B$

$x \in P(A \cap B)$

$\rightarrow$ Since x was arbitrary

$P(A) \cap P(B) \rightarrow P(A \cap B)$

Proved

$$b) A \subseteq B \text{ iff } P(A) \subseteq P(B)$$

$$A \subseteq B \rightarrow P(A) \subseteq P(B)$$

$$\forall x [x \in A \rightarrow x \in B] \text{ --- (1)} \quad \forall y [y \in P(A) \rightarrow y \in P(B)]$$

let y be arbitrary

$$y \in P(A)$$

$$y \subseteq A$$

$$y \in P(B)$$

$$y \subseteq B$$

$$\forall z (z \in y \rightarrow z \in A) \text{ --- (2)}$$

let z be arbitrary

$$\forall z (z \in y \rightarrow z \in B)$$

$$z \in y$$

$$\therefore z \in A \text{ --- (2)}$$

$$z \in B$$

$$\therefore z \in B \text{ --- (1)}$$

$$P(A) \subseteq P(A) \rightarrow A \subseteq B$$

$$\forall x [x \in P(A) \rightarrow x \in P(B)] \quad \forall y [y \in A \rightarrow y \in B]$$

$$\text{or } \forall x [x \subseteq A \rightarrow x \subseteq B]$$

let y be arbitrary

$$y \in A$$

$$y \in B$$

$$\forall x [\forall z (z \in x \rightarrow z \in A) \rightarrow \forall z (z \in x \rightarrow z \in B)]$$

By direct, we can't go anywhere, so let's try

by contradiction

$\therefore$ We will use the fact that every element is subset of itself.
 Suppose $P(A) \subseteq P(B)$. As $A \subseteq A$, $A \in P(A)$ & as $P(A) \subseteq P(B)$
 $A \in P(B)$ so $A \subseteq B$.

9) x is odd, y is odd

xy is odd

$$\exists k [x = 2k+1]$$

$$\exists m [ny = 2m+1]$$

$$\exists l [y = 2l+1]$$

$$x = 2k+1 \quad \text{existential not}$$

$$y = 2l+1$$

$$xy = (2k+1)(2l+1)$$

$$= 4kl + 2k + 2l + 1$$

$$= 2(2kl + k + l) + 1$$

$$\text{so } m = 2kl + k + l \text{ which } \in \mathbb{Z}.$$

10) x is odd
 y is odd

$x-y$ is even

$$\exists k [x-y = 2k]$$

$$\exists i [x = 2i+1]$$

$$\exists j [y = 2j+1]$$

$$x = 2i+1$$

$$y = 2j+1$$

$$x-y = 2i+1 - 2j-1$$

$$= 2i - 2j$$

$$= 2(i-j) \quad \text{so } k = i-j \in \mathbb{Z}.$$

11) n^3 is even iff n is even

n^3 is even $\longrightarrow$ n is even

$$\exists k [n^3 = 2k]$$

Suppose n is odd, we will prove that n^3 is odd by contraposition

$$\exists l [n = 2l + 1]$$

n^3 is odd

$$n = 2l + 1$$

$\therefore$ existential inst

$$\exists k [n^3 = 2k + 1]$$

$$n^3 = (2l)^3 + (1)^3 + 3(2l)^2(1) + 3(2l)(1)^2$$

$$= 8l^3 + 1 + 3(4l^2) + 6l$$

$$= 8l^3 + 12l^2 + 6l + 1$$

$$= 2[4l^3 + 6l^2 + 3l] + 1$$

$$\text{so } k = 4l^3 + 6l^2 + 3l$$

so n^3 is odd.

$$12) \forall x \in \mathbb{R} [\exists y \in \mathbb{R} (x + y = xy) \leftrightarrow x \neq 1]$$

let x be arbitrary

$$\exists y \in \mathbb{R} (x + y = xy) \leftrightarrow x \neq 1$$

$$\exists y \in \mathbb{R} (x + y = xy) \rightarrow x \neq 1$$

$$x + y_0 = xy_0 \quad \therefore \text{existential inst}$$

suppose $x = 1$

$$1 + y_0 = 1y_0$$

$$y_0 + 1 = y_0 \quad \therefore \text{contradiction, so } x \neq 1$$

$$x \neq 1 \rightarrow \exists y \in \mathbb{R} (x + y = xy)$$

$$\text{let } y = \frac{x}{x-1} \quad (\text{since } x \neq 1, \text{ so } x-1 \text{ is } \neq 0)$$

$$\begin{aligned} \text{then } x + y &= x + \frac{x}{x-1} = \frac{x(x-1) + x}{x-1} = \frac{x^2 - x + x}{x-1} \\ &= \frac{x^2}{x-1} = x \left(\frac{x}{x-1} \right) = xy \end{aligned}$$

$$14) \exists z \in R \forall x \in R^+ [\exists y \in R (y - x = y/x) \rightarrow x \neq z]$$

By solving $y - x = y/x$ for y , we get $y = \frac{-x^2}{1-x}$
 so x cannot be 1, since x cannot be 1, and $z \neq x$, we
 can let $z = 1$

let x be arbitrary.

$$\exists y \in R (y - x = y/x) \rightarrow x \neq z$$

Suppose $x = z$ so $x = 1$.

$$y - 1 = y/1$$

$$y - 1 = y \text{ which is contradiction so } x \neq z$$

$$x \neq z \rightarrow \exists y \in R (y - x = y/x)$$

$$\text{let } y = \frac{-x^2}{1-x}$$

$x \neq z$ so $x \neq 1$ so $1-x$ is defined.

$$y - x = \frac{-x^2}{1-x} - x = \frac{-x^2 - x(1-x)}{1-x}$$

$$= \frac{-x^2 - x + x^2}{1-x} = \frac{-x}{1-x} \left(\frac{x}{x} \right)$$

$$= \left(\frac{-x^2}{1-x} \right) \frac{1}{x}$$

$$= y \left(\frac{1}{x} \right) = y/x$$

$\therefore$ Mult Div by x

$$15) \cup \{A \setminus B \mid A \in F\} \subseteq \cup (F \setminus P(B))$$

$$\forall x [x \in \cup \{A \setminus B \mid A \in F\} \rightarrow x \in \cup (F \setminus P(B))]$$

let x be arbitrary

$$x \in \cup \{A \setminus B \mid A \in F\}$$

$$x \in \cup (F \setminus P(B))$$

$$\exists A \in F (x \in A \setminus B)$$

$$\exists D \in (F \setminus P(B)) (x \in D)$$

$$\exists A (A \in F \wedge x \in A \wedge x \notin B)$$

$$\exists D (D \in F \wedge D \notin P(B) \wedge x \in D)$$

$$A_0 \in F$$

$$x \in A_0 \text{ --- ①}$$

$$x \notin B \text{ --- ②}$$

let $D = A_0$. It fulfills all the conditions

$$A_0 \notin B \rightarrow \text{① ② (x now arbitrary)}$$

$$A_0 \notin P(B)$$

$$16) \forall A \in F \exists B \in G \wedge \exists x (x \in A \wedge x \in B)$$

$$\sim \exists x (x \in \cup F \wedge x \notin \cup G)$$

$$\forall A \in F \exists B \in G \wedge x (x \in A \rightarrow x \notin B) \text{ ①}$$

$$\forall x (x \in \cup F \rightarrow x \notin \cup G)$$

let x be arbitrary

$$x \in \cup F$$

$$x \notin \cup G$$

$$\exists A \in F (x \in A)$$

$$\sim \exists B \in G (x \in B)$$

$$\exists A (A \in F \wedge x \in A)$$

$$\forall B \in G (x \notin B)$$

$$= \forall B (B \in G \rightarrow x \notin B)$$

let B be arbitrary

$$B \in G \text{ --- ②}$$

$$x \notin B$$

$$A_0 \in F, x \in A_0 \text{ --- existential inst}$$

From ①, ②, ③, we conclude $x \notin B$.

$$17) A = \cup P(A)$$

$$A \subseteq \cup P(A) \quad \wedge \quad \cup P(A) \subseteq A$$

$$x \in A \rightarrow x \in \cup P(A) \quad \wedge \quad x \in \cup P(A) \rightarrow x \in A$$

$$x \in A \rightarrow x \in \cup P(A)$$

$$\exists B (B \in P(A) \wedge x \in B)$$

Any of these
2 approaches
can be used

Since $A \in P(A) \therefore A$ fact previously proved.

so let $B = A$

$$A \in P(A) \wedge x \in A \quad \text{proved}$$

Now

$$x \in \cup P(A) \rightarrow x \in A$$

$$\exists B (B \in P(A) \wedge x \in B) \quad x \in A$$

let $B = A$

$$A \in P(A) \wedge x \in A \quad \text{proved} \therefore \text{existential inst}$$

$$18) \cup (F \cap G) \subseteq (\cup F) \cap (\cup G)$$

$$\forall x \{ x \in \cup (F \cap G) \rightarrow x \in (\cup F) \cap (\cup G) \}$$

let x be arbitrary

$$x \in \cup (F \cap G)$$

$$x \in (\cup F) \cap (\cup G)$$

$$\exists A \in (F \cap G) (x \in A)$$

$$\exists A \in F (x \in A) \wedge \exists B \in G (x \in B)$$

$$\exists A (x \in F \wedge A \in G \wedge x \in A)$$

$$\exists A (x \in F \wedge x \in A)$$

$$A \in F \therefore \text{existential inst}$$

$$\exists B (x \in G \wedge x \in B)$$

$$A \in G$$

$$x \in A$$

Now we can prove both

$$19) (U \cap F) \cap (U \cap G) \subseteq U \cap (F \cap G) \text{ iff } \forall A \in F \forall B \in G (A \cap B \subseteq U \cap (F \cap G))$$

$$(U \cap F) \cap (U \cap G) \subseteq U \cap (F \cap G) \rightarrow \forall A \in F \forall B \in G (A \cap B \subseteq U \cap (F \cap G))$$

Let A, B be arbitrary sets of F, G respectively.
 $\forall x [x \in (U \cap F) \cap (U \cap G) \rightarrow x \in U \cap (F \cap G)]$ (5)

$$A \in F \rightarrow (1)$$

$$B \in G \rightarrow (2)$$

$$(A \cap B \subseteq U \cap (F \cap G))$$

$$\forall y (y \in A \cap B \rightarrow y \in U \cap (F \cap G))$$

Let y be arbitrary

$$y \in A \cap B$$

$$y \in A \quad (3)$$

$$y \in B \quad (4)$$

$$\text{From } (1) \& (3) \quad y \in U \cap F$$

$$\text{From } (2) \& (4) \quad y \in U \cap G$$

$$\text{so } y \in (U \cap F) \cap (U \cap G)$$

$$\text{From } (5) \quad y \in U \cap (F \cap G) \quad \text{Proved}$$

$$\forall A \in F \forall B \in G (A \cap B \subseteq U \cap (F \cap G))$$

$$\forall A \in F \forall B \in G \forall x [x \in A \cap B \rightarrow x \in U \cap (F \cap G)] \quad (U \cap F) \cap (U \cap G) \subseteq U \cap (F \cap G)$$

$$\forall x [x \in " \rightarrow x \in "]$$

Let x be arbitrary

$$x \in (U \cap F) \cap (U \cap G)$$

$$x \in U \cap (F \cap G)$$

$$x \in U \cap F \rightarrow \exists A \in F (x \in A) \quad (1)$$

$$x \in U \cap G \quad \exists B \in G (x \in B) \quad (2)$$

$$A \in F, x \in A \quad (1)$$

$$B \in G, x \in B \quad (2)$$

$$\text{so } x \in (A \cap B)$$

$$\text{From } (1) \& (2) \quad x \in U \cap (F \cap G) \quad \text{Proved.}$$

$$20) \neg \exists x [x \in U \cap F \wedge x \in U \cap G] \quad \forall A \in F \forall B \in G (A \cap B = \emptyset)$$

$$\forall x [x \in U \cap F \rightarrow x \notin U \cap G] \quad \text{①} \quad \forall A \in F \forall B \in G \forall y [y \in A \rightarrow y \notin B]$$

let A be arbitrary elmt of F

let B " " " " " G

let $y \in A$ be arbitrary

$$A \in F \quad \text{②}$$

$$y \notin B$$

$$B \in G \quad \text{③}$$

$$y \in A \quad \text{④}$$

We can prove by contradiction too
but let's use direct approach

$$y \in U \cap F \quad \text{②④}$$

$$y \notin U \cap G \quad \text{①}$$

$$\neg \exists x [x \in U \cap G \wedge x \in U \cap F]$$

$$= \neg \exists x [x \in U \cap G \wedge x \in U \cap F] = \forall x [x \notin U \cap G \vee x \notin U \cap F]$$

$$= \forall x [(x \in U \cap G \rightarrow x \notin U \cap F)] \quad \text{⑤}$$

$$y \notin B \quad \text{③ \& ⑤}$$

$$21) (U \cap F) \setminus (U \cap G) \subseteq U \cap (F \setminus G)$$

$$x \in U \cap F$$

$$x \in U \cap (F \setminus G)$$

$$x \notin U \cap G$$

$$\exists D (x \in F \cap D \wedge D \not\subseteq G \wedge x \in D)$$

$$\exists A \in F (x \in A)$$

$$\forall B \in G (x \notin B)$$

suppose $A \in G$, then $x \in G$ so $x \in U \cap G$ which is a contradiction
so $A \notin G$. let $D = A$. This proves

(b) $x \in A$ & $A \not\subseteq C$ is wrong to conclude $x \notin C$ since it means $\sim \exists A \in C (x \in A)$. What we proved here is $\exists A (A \in C \wedge x \in A)$ which are not same.

(c) $U(F|G) \subseteq (UF) \setminus (UG) : \text{iff } \forall A \in (F|G) \forall B \in G (A \cap B = \emptyset)$

let $A \in (F|G)$ be arbitrary
let $B \in G$ be arbitrary

$\forall x (x \in U(F G) \rightarrow x \in UF \setminus UG)$ ①	$A \cap B = \emptyset$
$A \in F G$	$\forall y (y \in A \rightarrow y \notin B)$
$B \in G$	let y be arbitrary
$y \in A$	$y \notin B$

Since $A \in F|G$ & $y \in A$

so $y \in U(F|G)$

From ① $y \in UF \setminus UG$

$y \in UF \wedge y \notin UG$

$\sim \exists C \in G (y \in C)$

$\forall C \in G (y \notin C)$

$\forall C (C \in G \rightarrow y \notin C)$

Since $B \in G$

so $y \notin B$ Proved -

$$(22) \quad U \neq V \rightarrow \exists A \in \mathcal{U} \exists B \in \mathcal{V} (A \not\subseteq B)$$

$$\exists x (x \in U \wedge x \notin V) \quad \exists A \in \mathcal{U} \exists B \in \mathcal{V} (y \in A \wedge y \notin B)$$

$$x_0 \in U \wedge x_0 \notin V$$

$$\exists A \in \mathcal{U} (x_0 \in A)$$

$$\sim \exists B \in \mathcal{V} (x_0 \in B)$$

$$\forall B \in \mathcal{V} (x_0 \notin B)$$

$$A_0 \in \mathcal{U}, x_0 \in A_0$$

Suppose $B \in \mathcal{V}$ is arbitrary
then $x_0 \notin B$

$$A = A_0$$

$$\forall B \in \mathcal{V} \exists y (y \in A \wedge y \notin B)$$

Arbitrary $B \in \mathcal{V}$

Let $A = A_0$, since B is arbitrary, so

$$\forall B \in \mathcal{V}, \text{ let } y = x_0, x_0 \in A \wedge x_0 \notin B.$$

Proved.

$$23) (b) \quad B \setminus (U_i \in \mathcal{I} A_i) = \bigcap_{i \in I} B \setminus A_i$$

$$\rightarrow x \in B \setminus (U_i \in \mathcal{I} A_i)$$

$$x \in B \quad \text{--- (1)}$$

$$x \notin U_i \in \mathcal{I} (A_i)$$

$$\sim \exists i \in I (x \in A_i)$$

$$\forall i (i \in I \rightarrow x \notin A_i)$$

$$x \in \bigcap_{i \in I} (B \setminus A_i)$$

$$\forall i \in I (x \in B \setminus A_i)$$

let $i \in I$ arbitrary

$$x \in B \wedge x \notin A_i$$

By ~~modus ponens~~ modus ponens

By ~~modus ponens~~ modus ponens
 $x \notin A_i$

$$\text{also } x \in B \quad \text{--- (1)}$$

Proved

$$\leftarrow x \in \bigcap_{i \in I} (B \setminus A_i)$$

$$\forall i \in I (x \in B \setminus A_i)$$

$$\forall i (i \in I \rightarrow x \in B \setminus A_i)$$

We can apply universal inst
suppose $i_0 \in I$

$$\text{th } x \in B \text{ \& } x \notin A_{i_0}$$

so it is proved

$$x \in B \setminus (\bigcup_{i \in I} A_i)$$

$$x \in B$$

$$x \notin \bigcup_{i \in I} A_i$$

$$\sim \forall i \in I (x \in A_i)$$

$$\exists i \in I (x \notin A_i)$$

$$\exists i (i \in I \wedge x \notin A_i)$$

$$(c) B \setminus (\bigcap_{i \in I} A_i) = \bigcup_{i \in I} (B \setminus A_i)$$

$$\rightarrow x \in B \setminus (\bigcap_{i \in I} A_i)$$

$$x \in B \wedge x \notin \bigcap_{i \in I} A_i$$

$$\forall i (i \in I \rightarrow x \notin A_i)$$

Using universal inst

$$i_0 \in I \quad x \notin A_{i_0}$$

$$\text{Also } x \in B$$

Proved.

$$\leftarrow \bigcup_{i \in I} (B \setminus A_i)$$

$$\exists i (i \in I \wedge x \in B \setminus A_i)$$

$$i_0 \in I \quad x \in B \quad x \notin A_{i_0}$$

~~suppose~~

proved

$$B \setminus (\bigcap_{i \in I} A_i)$$

$$\bullet x \in B$$

$$\bullet \forall i (i \in I \rightarrow x \notin A_i)$$

~~let it be arbitrary~~

$$\cancel{x \notin A_i}$$

$$x \notin \bigcap_{i \in I} A_i$$

$$\sim \forall i \in I (x \in A_i)$$

$$\exists i \in I (x \notin A_i)$$

24) $\forall x (x \in \bigcup_{i \in I} (A_i \setminus B_i) \rightarrow x \in (\bigcup_{i \in I} A_i) \setminus (\bigcap_{i \in I} B_i)$

(a)

let x be arbitrary

suppose $x \in \bigcup_{i \in I} (A_i \setminus B_i)$

$x \in (\bigcup_{i \in I} A_i) \setminus (\bigcap_{i \in I} B_i)$

$\exists i \in I (x \in A_i \setminus B_i)$

$x \in \bigcup_{i \in I} A_i$

$\exists i (i \in I \wedge x \in A_i \wedge x \notin B_i)$

$x \notin \bigcap_{i \in I} B_i$

$\exists i \in I, x \in A_i, x \notin B_i$

$\exists i \in I (x \in A_i)$

$\exists i \in I (x \notin B_i)$

Proved.

~~$\exists i \in I$~~

(b) $A_1 = \{1, 2, 3\}$ $A_2 = \{2, 3\}$

$B_1 = \{2, 3\}$ $B_2 = \{2, 5\}$

25) $\forall x [x \in \bigcup_{i \in I} (A_i \cap B_i) \rightarrow x \in (\bigcup_{i \in I} A_i) \cap (\bigcup_{i \in I} B_i)]$

let x be arbitrary

suppose $x \in \bigcup_{i \in I} (A_i \cap B_i)$

$x \in (\bigcup_{i \in I} A_i) \cap (\bigcup_{i \in I} B_i)$

$\exists i \in I (x \in A_i \cap B_i)$

$x \in \bigcup_{i \in I} A_i \wedge x \in \bigcup_{i \in I} B_i$

$\exists i (i \in I \wedge x \in A_i \wedge x \in B_i)$

$\exists i \in I (x \in A_i)$

$\exists i \in I, x \in A_i, x \in B_i$

$\exists i \in I (x \in B_i)$

Proved.

(b) $A_1 = \{1, 2, 3\}$ $A_2 = \{3, 4\}$

$B_1 = \{2, 3\}$ $B_2 = \{1, 4\}$

$$26) \forall a \forall b \exists c (a|c \wedge b|c)$$

suppose $a \in \mathbb{Z}$ and $b \in \mathbb{Z}$ be arbitrary

$$\exists c (a|c \wedge b|c)$$

$$\text{let } c = ab$$

as $a \in \mathbb{Z}$ & $b \in \mathbb{Z}$, $ab \in \mathbb{Z}$

$$\exists c (an = c \wedge bm = c) \\ \text{where } n, m \in \mathbb{Z}$$

$$\text{since } a(b) = c \text{ so } a|c$$

$$\& \text{ since } b(a) = c \text{ so } b|c \quad \text{Proved.}$$

$$27) (a) \forall n \in \mathbb{Z}^+ [15|n \rightarrow (3|n \wedge 5|n)]$$

let $n \in \mathbb{Z}$ be arbitrary

$$\rightarrow 15|n$$

$$3|n \wedge 5|n$$

$$\exists k \in \mathbb{Z} (15k = n)$$

$$\exists k \in \mathbb{Z} (3k = n)$$

$$15k = n \quad (\because \text{existential inst})$$

$$\exists l \in \mathbb{Z} (5l = n)$$

$$3 \times (5k) = n \text{ so } 3|n \quad (\text{let } k = 5m)$$

$$\& 5 \times (3k) = n \text{ so } 5|n \quad (\text{let } l = 3m)$$

$$\leftarrow \text{suppose } 3|n \& 5|n$$

$$15|n$$

$$\exists k \in \mathbb{Z} (3k = n)$$

$$\exists m \in \mathbb{Z} (15m = n)$$

$$\exists l \in \mathbb{Z} (5l = n)$$

$$3k' = n \quad \text{universal inst}$$

$$5l' = n$$

$$2l' - k'$$

suppose $n = \text{any}$ then clearly $2l' - k'$ is an integer

$$15(2l' - k') = 2 \times 3 \times 5l' - 5 \times 3k'$$

$$= 6 \times 5l' - 5 \times 3k'$$

$$= 6n - 5n = n \quad \text{Proved}$$

$$(q) \sim \forall n [60|n \leftrightarrow (6|n \wedge 10|n)]$$

$$\exists n [(60|n \rightarrow (6|n \wedge 10|n)) \wedge ((6|n \wedge 10|n) \rightarrow 60|n)]$$

$$\exists n [(60 \nmid n \vee (6|n \wedge 10|n)) \wedge (\sim(6|n \wedge 10|n) \vee 60|n)]$$

we will prove by contradiction. Suppose $\sim p$.

$$\text{i.e. suppose } \forall n [60|n \rightarrow (6|n \wedge 10|n) \wedge ((6|n \wedge 10|n) \rightarrow 60|n)]$$

then n is arbitrary & we have 2

$$60|n \rightarrow (6|n \wedge 10|n)$$

$$(6|n \wedge 10|n) \rightarrow 60|n$$

Suppose $n = 30$, then $6|n$ & $10|n$ but $60|n$

so that's contradiction.

↳ Basically we simply derived a contradiction to prove that the given statement is True. we provided an example because we only had to provide 1.

If was a goal of prove $p \wedge q$

$p \wedge q$

Now if we suppose $\neg(p \wedge q)$ and make a contradiction our goal we have

given

goal

$$\neg(p \wedge q)$$

contradiction

$$\sim p \vee \sim q$$

Since this is OK, we can derive contradiction from any one of these 2.